University of Geneva

Faculty of Science

My Course

Digital Forensics

14x065

Michel Jean Joseph Donnet

2026-03-04

Contents

1 Introduction

A large part of the operations performed on an image in digital processing are carried out in the Fourier domain. Working in the Fourier domain speeds up image filtering and expands the range of possibilities by providing a different representation of the image.

The Fourier transform, which enables the transition between the spatial domain and the Fourier domain, transforms real numbers into complex numbers.

According to Euler's formula, a complex number can be represented as a magnitude and a phase: $z = |M| \ e^{j\varphi}$, where $|M| \in \mathbb{R}$ is the magnitude and $\varphi \in [0, 2\pi)$ is the phase.

In image processing, the magnitude of a given image I is given by the absolute value of the Fourier transform $|F(I)|$, while the phase is given by the angle of the Fourier transform $\angle F(I)$.

In this work, we will attempt to analyze the impact of the amplitude and phase of the Fourier transform by replacing and mixing the phases of two images.

2 Methodology

3 Implementation

4 Results

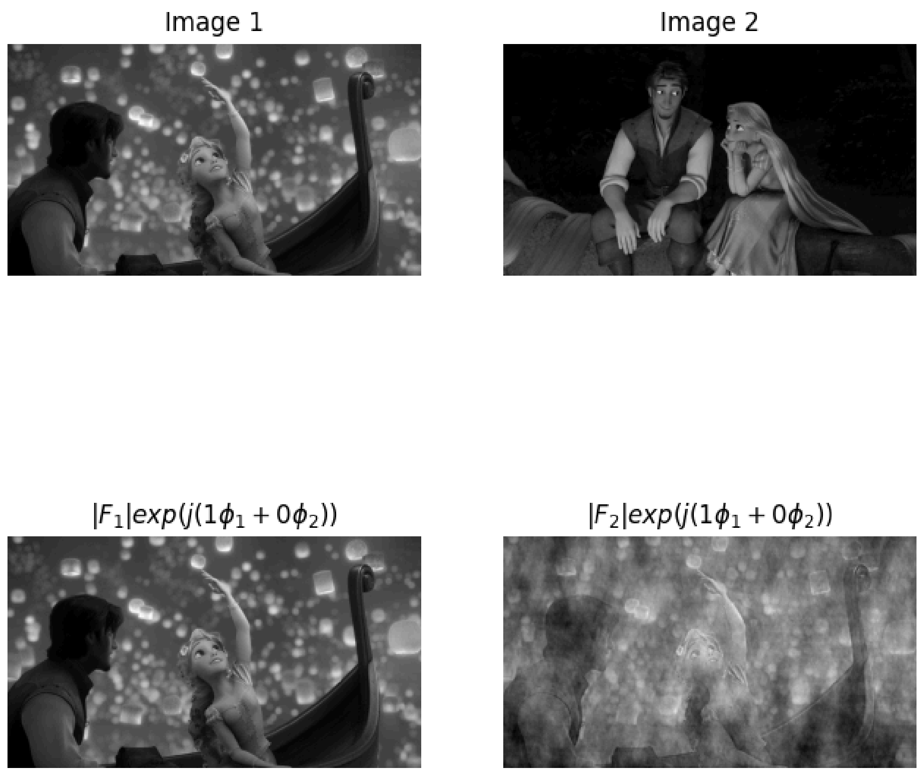

Figure 1: Phase mixture with $\varphi = \varphi_{\text{img\_1}}$

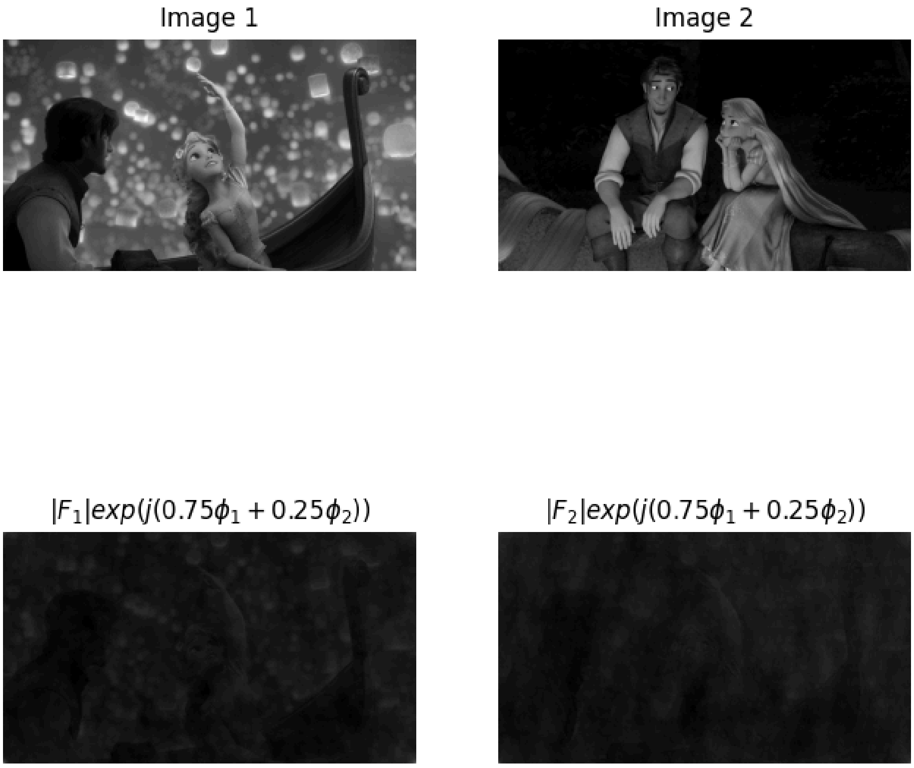

Figure 2: Phase mixture with $\varphi = \frac{3}{4}\varphi_{\text{img\_1}} + \frac{1}{4}\varphi_{\text{img\_2}}$

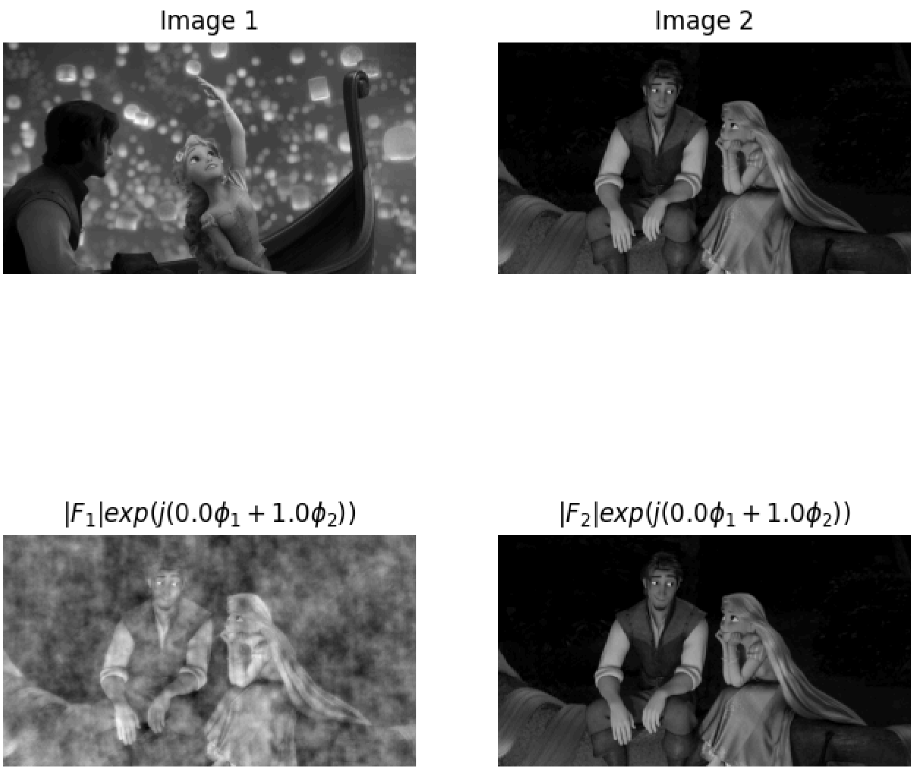

Figure 3: Phase mixture with $\varphi = \varphi_{\text{img\_2}}$

5 Discussion

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6 Conclusion